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B. E. (Fourth Semester) Examination, April-May, 2016

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL MEASUREMENTS & MEASURING INSTRUMENTS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt Part (a) of each unit is compulsory and any two from parts (b), (c) and (d).

Unit-I

1. (a) What is meant by low, medium and high resistance? 2

(b) Explain the construction and principle of operation of Kelvin double bridge. Mention why these bridge is used for accurate measurement of low resistance. 7

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(c) Describe and explain with the help of neat sketch the construction of working of megger. 7

(d) 3 resistance have to following ratings

$R_1 = 2000\, r \pm 5\%$, $R_2 = 100\, r \pm 5\%$

$R_3 = 50\, r \pm 5\%$

Determine the magnitude of resultant resistance and limiting error in percentage and ohms, if the above resistances are connected :

(i) Series

(ii) Parallel 7

Unit-II

2. (a) Name the commonly used detection for ac bridge. 2

(b) Explain Wagner's earthing device to minimize stray capacitance by ac bridge. 7

(c) Describe how schering bridge can be used for measurement of an unknown capacitance and its loss angle. Derive the conditions of balance and draw phasor diagram of the bridge circuit under balanced condition. 7

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(d) List the factor that may lead to in accuracy in measurement by ac bridge. The four arm of bridge are :

arm AB : an imperfect capacitance C_1 with an equivalent series resistacne r_1

arm BC : an non inductive resistance R_3

arm CD : a non inductive resistance R_4.

arm DA : an imperfect capacitance C_2 with an equivalent resistance r_2 in series with a resistance R_2.

A supply of 450 Hz is given between terminal A and C of a detection is connected between B and D.

At balance condition $R_2 = 5\,\Omega$, R_3 2000 Ω,

$R_4 = 2950\,\Omega$, $C_2 = 0{\cdot}5\,\mu F$ and $r_2 = 0{\cdot}4\,\Omega$.

Calculate the value of C_1 and r_1 of also the dissipating factor for this capacitor.

Unit-III

(a) Name the most commonly used galvanometer. 2

(b) Explain the intrinsic constants of a D'Arsonval galvanometer and write down the equation for motion

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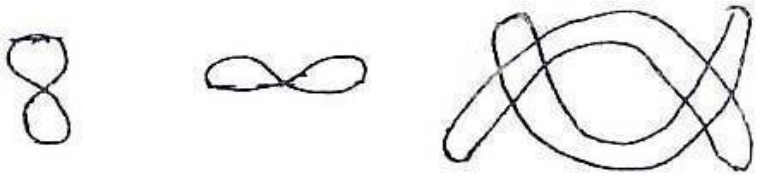
of a galvanometers. Explain the response of galvanometer under damped conditions. 7

(c) Draw the block diagram of oscilloscope and explain its main parts. 7

(d) Calculate the ratio of vertical and horizontal frequencies for an oscilloscope which display the following Lissajous figure shown below. 7



(i) (ii) (iii)

Unit-IV

4. (a) Classify measuring instrument. 2

(b) Explain the construction and working of PMMC instrument. State its advantage. 7

(c) Explain single phase electro dynamometer power factor meter. 7

(d) Explain the construction and working of ratiometer types frequency meter. 7

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Unit-V

5. (a) How the error due to resistance of pressure coil is compensated automatically in a dynamometer type wattmeters? 2

(b) What are the errors in dynamometer type wattmeter and how they are compensated? 7

(c) Describe the construction and working of single phase induction type energy meter. 7

(d) The power flowing in a 3-ϕ, 3 wire balanced load system is measured by two wattmeter method. The reading of wattmeter A is 7500 W and of wattmeter B is – 1500 W.

(i) What is the power factor of the system?

(ii) If the voltage of the circuit is 400 V, what is the value of capacitance which must be introduced in each phase to cause the whole of the power measured to appear on wattmeter A. The frequency is 50 Hz. 7

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